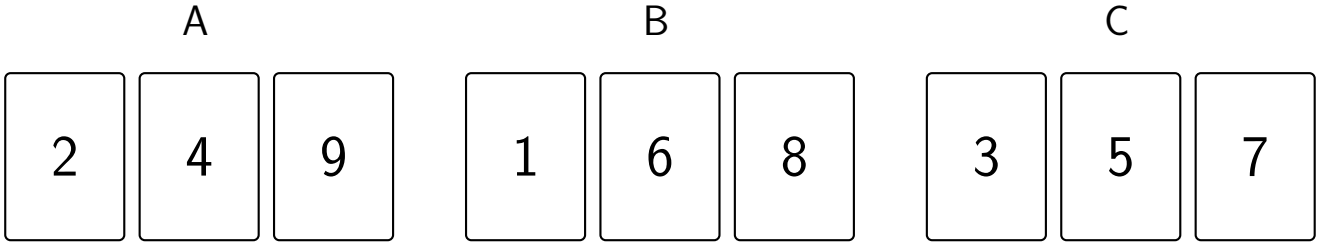
Each physical card is equally likely. Mix each deck separately and draw one card from each. Return both cards before the next round. The larger number wins. The two decks in a comparison must not share a number. Card order in a deck does not matter.

Problem 1: For each pair of decks, find how many of the possible card pairs each deck wins. Is there a deck that wins more than half its pairs against both others? Do the three deck totals settle this question?



Problem 2: Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

